

Seat No./Enrollment No.: _____

GUJARAT TECHNOLOGICAL UNIVERSITY
BACHELOR OF ENGINEERING – SEMESTER 7 – EXAMINATION – SUMMER 2026

Subject Code: 3174201

Date: 20/05/2026

Subject Name: Deep Learning: Principles and Practices

Time: 02:30 PM TO 05:00 PM

Total Marks: 70

Instructions:

- 1. Attempt all questions.**
- 2. Make suitable assumptions wherever necessary.**
- 3. Figures to the right indicate full marks.**
- 4. Simple and non-programmable scientific calculators are allowed.**

- Q.1**
- (a) List Applications of deep learning. **03**
 - (b) Explain advantages and disadvantages of deep learning. **04**
 - (c) Illustrate Neural Network concept with reference to deep learning. **07**
- Q.2**
- (a) List Applications of ANN. **03**
 - (b) Explain Deep Feed forward Network. **04**
 - (c) Illustrate the Artificial Neural Network architecture with diagram. **07**
- OR
- Q.2**
- (c) Analyze activation function with its types of Neuron activation function. **07**
- Q.3**
- (a) List fuzzy logic operators in deep learning. **03**
 - (b) Summarize applications of Neural Network Driven Fuzzy Reasoning. **04**
 - (c) Illustrate fuzzy inference with fuzzy rule based system having fuzzy logic and defuzzification. **07**
- OR
- Q.3**
- (a) List advantages of Neuro fuzzy controller. **03**
 - (b) Explain Network driven fuzzy reasoning and neural fuzzy relational system. **04**
 - (c) Illustrate recent application of fuzzy logic in industry. **07**
- Q.4**
- (a) List basic features of tensorflow. **03**
 - (b) Explain optimization in Deep learning. **04**
 - (c) Illustrate tensorflow workflow in detail. **07**
- OR
- Q.4**
- (a) Identify why transfer learning important. **03**
 - (b) Explain natural language processing with its basics. **04**
 - (c) Illustrate Transfer Learning with MobileNetV2 for MNIST Classification. **07**
- Q.5**
- (a) Define Deep Learning Algorithms. **03**
 - (b) Compare ANN and BNN. **04**

- (c) Illustrate Kohonen Self Organizing Map (KSOM) architecture, including its input layer, weight vectors, and clusters. **07**

OR

- Q.5**
- (a) List Generative Adversarial Network (GAN) algorithm steps. **03**
- (b) Compare Radial Basis Function Network (RBFN) and MLP. **04**
- (c) Illustrate Deep Belief Network (DBN) system working in detail. **07**
